

JEE MAIN REVISION MASTER: BIOMOLECULES

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 198 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Amino Acids, Peptide Bonds & Nucleic Acids	104 PYQs	Proteins & DNA
Very High	Carbohydrates (Glucose, Fructose & Glycosidic Linkage)	100 PYQs	Carbohydrates

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Amino Acids, Peptide Bonds & Nucleic Acids	$\text{Peptide linkage: } -\text{CO-NH}-, \quad \text{Zwitterion: } \text{H}_3\text{N}^+-\text{CHR}-\text{COO}^-, \quad \text{DNA: } \text{A}=\text{T}, \text{G}\equiv\text{C}$
Carbohydrates (Glucose, Fructose & Glycosidic Linkage)	$\text{Sucrose (non-reducing)} \xrightarrow{\text{hydrolysis}} \text{D-(+)-glucose} + \text{D-(-)-fructose} \quad \text{Invert sugar, levorotatory}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Amino Acids, Peptide Bonds & Nucleic Acids (Very High Yield — 104 PYQs)

Core Principle: Essential amino acids, zwitterionic dipolar character, protein denaturation, double-helix structure of DNA, and hydrogen-bonded base pairs.

■ **Caution / JEE Pitfall:** Denaturation destroys secondary, tertiary, and quaternary protein structures, but leaves the primary sequence (peptide bonds) INTACT.

■ **Speed Trick:** All naturally occurring α -amino acids are optically active EXCEPT Glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$), which is achiral.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Match the List-I with List-II List-I Name of amino acid List-II One letter symbol/type A. Arginine I. D/Non-essential B. Aspartic acid II. R..." [Ans: A]
- [2026 Morning Shift] Q#1 (MCQ): "In the given pentapeptide, find out an essential amino acid (Y) and the sequence present in the pentapeptide : $\text{Choose the correct a...}$ " [Ans: N/A]
- [2026 Evening Shift] Q#2 (MCQ): "Identify the correct pair having amino acid (A) and the hormone (B) that is iodinated derivative of the amino acid (A). (T and Y represent o..." [Ans: N/A]

3.2 Carbohydrates (Glucose, Fructose & Glycosidic Linkage) (Very High Yield — 100 PYQs)

Core Principle: Structure of D-glucose and D-fructose, cyclic hemiacetals, anomeric carbons (α/β), reducing vs non-reducing sugars, and invert sugar.

■ **Caution / JEE Pitfall:** Sucrose is a NON-REDUCING sugar because both reducing groups (hemiacetal -OH of glucose and hemiketal -OH of fructose) are tied in glycosidic linkage.

■ **Speed Trick:** Hydrolysis of dextrorotatory sucrose produces a levorotatory mixture because fructose has a larger specific rotation (-92.4°) than glucose ($+52.5^\circ$).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "The incorrect statement from the following with respect to carbohydrates is :..." [Ans: N/A]
- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : and are two anomers of D-(+)-glucose. Statement II : The open chain forms o..." [Ans: N/A]
- **[2026 Morning Shift] Q#1 (MCQ):** "Given below are two statements : Statement I : Sucrose is dextrorotatory. However, sucrose upon hydrolysis gives a solution having mixture o..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Morning Shift (Q#1)

Match the List-I with List-II List-I Name of amino acid List-II One letter symbol/type A. Arginine I. D/Non-essential B. Aspartic acid II. R/Essential C. Lysine III. E/Non-essential D. Glutamic acid IV. K/Essential Choose the correct answer from the options given below :

- (A) -II, B-I, C-IV, D-III [Correct]
- (B) -IV, B-III, C-II, D-I
- (C) -III, B-IV, C-I, D-II
- (D) -II, B-IV, C-I, D-III

Correct Answer: A

Solution Analysis: Match each amino acid with its correct one-letter symbol and whether it is essential or non-essential: Arginine $\rightarrow$ one-letter symbol is R In standard school-level classification here, it is matched as Essential So, A $\rightarrow$ II Aspartic acid $\rightarrow$ one-letter symbol is D It is Non-essential So, B $\rightarrow$ I Lysine $\rightarrow$ one-letter symbol is K It is Essential So, C $\rightarrow$ IV Glutamic acid $\rightarrow$ one-letter symbol is E It is Non-essential So, D $\rightarrow$ III Thus the correct matching is: A-II, B-I, C-IV, D-III So, the correct option is

Benchmark Problem 2: JEE Main 2026 (Online) 6th April Morning Shift (Q#2)

Match the List-I with List-II
List-I
Deficiency Disease
List-II
Vitamin
A. Scurvy
I. Pyridoxine
B. Convulsions
II. Vitamin A
C. Cheilosis
III. Ascorbic Acid
D. Xerophthalmia
IV. Riboflavin
Choose the correct answer from the options given below :

- (A) -I, B-III, C-II, D-IV
- (B) -I, B-III, C-IV, D-II
- (C) -III, B-I, C-IV, D-II **[Correct]**
- (D) -III, B-I, C-II, D-IV

Correct Answer: C

Solution Analysis: Let us recall the vitamin deficiencies and the corresponding diseases: Scurvy – caused due to deficiency of Vitamin C (Ascorbic acid) . → So, A → III Convulsions – occur due to deficiency of Vitamin B₆ (Pyridoxine). → So, B → I Cheilosis – characterized by cracks at the corners of the mouth; caused by deficiency of Vitamin B₂ (Riboflavin). → So, C → IV Xerophthalmia – dryness of the cornea and conjunctiva; caused by deficiency of Vitamin A . → So, D → II Hence, the correct matching is: A-III,B-I,C-IV,D-II Therefore, the correct